

Theorem. (Erdős #12(i)) There is an infinite set $A \subseteq \mathbb{N}$ satisfying that there are no distinct $a, b, c \in A$ with $a < b, c$ satisfying $a|b + c$ and

$$\liminf_{N \rightarrow \infty} \frac{|A \cap \{1, \dots, N\}|}{N^{1/2}} > 0.$$

Proof. We will construct a sequence of "blocks" that individually cannot contain such a, b, c , and with a careful choice of parameters, their union forms the desired set. By precisely controlling the growth and modular residues of these blocks, we ensure the set remains dense enough to satisfy the $\liminf$ condition while avoiding all forbidden divisibility relations.

Let $f : \mathbb{N} \rightarrow \mathbb{N}$ be defined by $f(0) = 0$ and $f(n) = 3f(\lfloor n/2 \rfloor) + (n \bmod 2)$. The function f takes a number represented in base 2 and outputs the number given by the same representation but read in base 3. In particular, the base 3 representation of $f(n)$ will contain only 0s and 1s. For this reason, $f(a) + f(b) = 2f(c)$ implies $a = b = c$, ensuring the sequence $(f(n))_{n \in \mathbb{N}}$ contains no 3-APs.

Next, let F_i be the i -th odd prime, and write $M_i = \prod_{j < i} F_j$ and $V_i = F_i M_i$. By Bertrand's postulate, we have $F_i \leq 2^{i+2}$. By the Chinese Remainder Theorem, for each i , let $C_i < V_i$ be the unique integer such that $C_i \equiv 0 \pmod{F_i}$ and $C_i \equiv 1 \pmod{M_i}$.

Now we define the parameters for the blocks (B_i) . Let

- $Y_i = 3^{(i+20)^3}$ be the exponential bounding scale to control the distance between blocks.
- $P_i = 10V_i Y_i + C_i$ be the starting coordinate of block i .
- $X_i = \lfloor \sqrt{P_{i+1}} \rfloor$ be the element capacity of block i . Note that the capacity of B_i is directly tied to the starting point of the next block B_{i+1} .

Formally, define $B_i = \{P_i + V_i f(y) \mid y < X_i\}$. The purpose of Y_i is to keep the blocks relatively narrow while ensuring they are well-spaced. Note that if $y < 2^k$, then $f(y) < 3^k$. Letting $k = (i+20)^3$, we want to guarantee that for any $y < X_i$, we have $f(y) < Y_i = 3^k$. To do this, it suffices to show $X_i \leq \sqrt{P_{i+1}} < 2^k$, which is equivalent to $P_{i+1} \leq 4^k$.

Observe that $P_{i+1} \leq 11V_{i+1}Y_{i+1} \leq 11 \cdot 2^{(i+4)^2} \cdot 3^{(i+21)^3}$, following from the bound on F_i . This is asymptotically strictly smaller than $4^k = 4^{(i+20)^3}$. By adding the shift of 20, the inequality $11 \cdot 2^{(i+4)^2} \cdot 3^{(i+21)^3} \leq 4^{(i+20)^3}$ holds universally for all $i \geq 1$.

Set $A = \bigcup_{i \in \mathbb{N}} B_i$. Clearly, A is an infinite set. Suppose, for the sake of contradiction, that there exist distinct $a, b, c \in A$ with $a < b, c$ such that $a \mid (b + c)$. Let $a \in B_i$. We analyze two cases:

- (Case 1: Cross-block) Suppose at least one of b, c is not in B_i . Without loss of generality, assume $b \in B_j$ for some $j > i$ (since $b > a$). Since $a \in B_i$, we have $a \equiv 0 \pmod{F_i}$, and thus $b + c \equiv 0 \pmod{F_i}$ by the divisibility condition. However, by our CRT construction, $b \equiv 1 \pmod{F_i}$. The element c must belong to some block B_m with $m \geq i$, so $c \equiv 0 \pmod{F_i}$ (if $m = i$) or $c \equiv 1 \pmod{F_i}$ (if $m > i$). Therefore, $b + c \equiv 1 \pmod{F_i}$ or $b + c \equiv 2 \pmod{F_i}$. Since $F_i \geq 3$, this directly contradicts $b + c \equiv 0 \pmod{F_i}$.
- (Case 2: Same block) Suppose $b, c \in B_i$. They must be tightly clustered around P_i . Because $y < X_i$, we have $f(y) < Y_i$. The maximum "noise" added to any block element is $V_i Y_i$. Since $P_i > 10V_i Y_i$, any $x \in B_i$ must satisfy $x \in [P_i, 1.1P_i]$. The divisibility condition $a \mid (b + c)$ implies $ma = b + c$ for some integer m . Because $b + c \in [2P_i, 2.2P_i]$ and $a \in [P_i, 1.1P_i]$, it must be that $m = 2$, implying $2a = b + c$.

Writing the elements as $a = P_i + V_i f(a')$, $b = P_i + V_i f(b')$, and $c = P_i + V_i f(c')$, substitution and cancellation yield $2f(a') = f(b') + f(c')$. This is impossible for distinct a', b', c' because the image of f avoids 3-APs, contradicting the assumption that a, b, c are distinct.

Finally, we show that $C(N) = |A \cap \{1, \dots, N\}| \geq c\sqrt{N}$ for some constant $c > 0$. We split into two cases:

1. N falls in the gap between B_i and B_{i+1} ; precisely, $P_i + V_i Y_i \leq N < P_{i+1}$. Here, $C(N)$ counts at least all elements in B_i , meaning $C(N) \geq X_i = \lfloor \sqrt{P_{i+1}} \rfloor$. Since $N \leq P_{i+1}$, we have $\sqrt{N} \leq \sqrt{P_{i+1}}$. Thus, $\frac{C(N)}{\sqrt{N}} \geq \frac{\lfloor \sqrt{P_{i+1}} \rfloor}{\sqrt{P_{i+1}}} \approx 1$.
2. N is inside the block B_i ; precisely, $P_i \leq N < P_i + V_i Y_i$. A pessimistic lower bound counts only the elements up to the previous block, giving $C(N) \geq X_{i-1} = \lfloor \sqrt{P_i} \rfloor$. Observe that $V_i Y_i \leq 0.1P_i$, so $N \leq 1.1P_i < 2P_i$. Consequently, $\sqrt{P_i} > \sqrt{N/2}$, which implies $\frac{C(N)}{\sqrt{N}} \geq \frac{1}{\sqrt{2}}$.

In both cases, $\liminf_{N \rightarrow \infty} \frac{C(N)}{N^{1/2}} \geq \frac{1}{\sqrt{2}} > 0$. □